

Question Paper

Exam Date & Time: 31-Dec-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
FIFTH SEMESTER B.TECH. DEGREE, MAKEUP EXAMINATIONS - DECEMBER 2025
WEB AND SOCIAL MEDIA ANALYTICS [DS_3153]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Analyse which region is most at risk of vaccine spoilage in the next quarter, with justification based on the given data. (5)

A) Suggest three prescriptive actions to minimize spoilage and improve cold-chain reliability.

Parameter	Region A	Region B	Region C
Average Temperature Deviation (°C)	1.2	3.8	2.0
Number of Compressor Failures	2	9	3
Door Seal Alerts (per month)	1	6	2
Product Spoilage Rate (%)	0.5	4.2	1.1

Course Outcome: "CO2" / Cognitive Level: "Analysis"

- B) A retail company wants to improve its sales forecasting accuracy. They currently collect large volumes of sales data but face difficulty in generating useful insights for future planning. Apply your understanding of Business Intelligence (BI) and Decision Support Systems (DSS) to explain with diagram how these tools can help the company make better sales forecasts. (3)

Course Outcome: "CO2" / Cognitive Level: "Application"

- C) Uber plans to launch a "Green Ride" option that matches passengers with electric vehicles (EVs). Articulate how Big Data analytics can be applied to ensure efficient EV allocation and route planning. (2)

Course Outcome: "CO2" / Cognitive Level: "Application"

- 2) A large e-commerce company wants to analyze thousands of customer reviews to identify recurring product issues and satisfaction levels. Using the sample review: (5)

A) **"The delivery was late, but the product quality was excellent."**

Apply and demonstrate the main NLP preprocessing steps - tokenization, stop word removal, lemmatization/stemming, and POS tagging - using Python code (use any conceptual library). Based on the output, briefly explain how these steps transform the raw text into structured data suitable for Text Mining.

Course Outcome: "CO1" / Cognitive Level: "Application"

- B) Using the processed text from Q2a, write a short Python code that constructs a simple Term-Document Matrix (TDM). (3)
Explain how NLP preprocessing reduces noise and dimensionality, helping improve the efficiency of this matrix for downstream Text Mining tasks such as clustering or keyword extraction.

Course Outcome: "CO1" / Cognitive Level: "Application"

- C) Explain how sentiment analysis can help the company evaluate overall customer satisfaction from online reviews. (2)

Course Outcome: "CO1" / Cognitive Level: "Understand"

- 3) An e-commerce company wants to analyse customer reviews for its new range of smartphones. The reviews are collected in textual form from multiple online sources. The goal is to determine customer sentiment (positive, negative, neutral) and identify key factors influencing satisfaction (e.g., battery, camera, price). (5)
A) Compare the use of lexicon-based and machine learning-based approaches for analyzing these reviews.
Suggest which text preprocessing techniques (e.g., tokenization, stopword removal, stemming/lemmatization) are most essential.

Course Outcome: "CO1" / Cognitive Level: "Application"

- B) Choose the most popular application areas for sentiment analysis and how can sentiment analysis be used for brand management? (3)

Course Outcome: "CO1" / Cognitive Level: "Analysis"

- C) Describe the acoustic approach to speech analytics and linguistic approach to speech analytics. (2)

Course Outcome: "CO1" / Cognitive Level: "Understand"

- 4) A digital marketer attempts to optimize a major e-commerce website for usability and conversion solely by analyzing raw Web Usage Mining (WUM) data, such as Clickstream Analysis derived primarily from Web Server Logs. (5)
A) Analyze the deficiencies of this approach in diagnosing the root causes of poor performance metrics like a high Bounce Rate or low Conversion Rate. Justify why Web Site Optimization requires convergent validation by correlating these WUM quantitative findings (the "what") with Voice of Customer (VOC) data (the "why"), and explain how incorporating this qualitative input is essential to move beyond merely descriptive metrics towards effective management and problem resolution

Course Outcome: "CO2" / Cognitive Level: "Analysis"

- B) Write a Python program that simulates a Google Analytics (GA4) traffic and conversion analysis system. (3)
Your program should:
1. Store data for at least four different traffic sources (for example: Organic Search, Paid Search, Direct, Referral) each source should include:
 - total users
 - total conversions
 - country
 - device type
2. Calculate the conversion rate (%) for each traffic source.

3. Identify and display which traffic source has the highest conversion rate.
4. Segment conversions by country and by device type.

Course Outcome: "CO3" / Cognitive Level: "Application"

- C) Write a Python program that analyses a list of customer tweets and prints "Positive Brand Image" if the average sentiment polarity is greater than 0, else prints "Negative Brand Image" (2)

Course Outcome: "CO3" / Cognitive Level: "Analysis"

- 5) The Smart City Planning Committee of "GreenVille" aims to select the most suitable renewable energy source to meet its sustainability goals. The committee wants a decision-making approach that considers multiple factors beyond cost alone. (5)

- A) They shortlist four alternatives:

1. Solar Power
2. Wind Power
3. Hydropower
4. Biomass Energy

After initial discussions, the committee identifies four key criteria to evaluate each alternative:

- C1: Installation Cost
- C2: Energy Efficiency
- C3: Environmental Impact
- C4: Maintenance Requirements

The committee decides to apply the Analytic Hierarchy Process (AHP) to structure the decision and determine the best alternative.

Analyse how the AHP process can be applied to this problem. Clearly outline the steps the committee should follow, from defining the goal to ranking the alternatives.

Course Outcome: "CO4" / Cognitive Level: "Application"

- B) A start-up company, *GreenTech Solutions*, manufactures eco-friendly electric scooters. The management is planning to expand production and needs to forecast profitability for the next financial year. (3)

Key parameters influencing profit include:

- Selling price per scooter
- Cost of raw materials
- Labor cost
- Monthly demand (sales volume) Classify the following: *What if the cost of batteries rises by 10%?*
- *What if the demand increases by 20% due to government subsidies?*
- *What if the selling price is reduced to attract more customers?*

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- C) A retail company uses a spreadsheet model to plan monthly profits. The manager wants to find out what level of advertising spend is needed to reach a target profit of ₹1,00,000. (2)
- Using this case, explain the core function of Goal Seeking in decision Modeling and describe, in simple terms, how it determines the required input value to reach a desired outcome.

Course Outcome: "CO4" / Cognitive Level: "Understand"

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Event Name	Subject
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#	Course Outcome	Description	Marks
1	1	CO 1	20
2	2	CO 2	15
3	3	CO 3	5
4	4	CO 4	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026

DS_3153:WEB AND SOCIAL MEDIA
ANALYTICS

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	3	6(12%)
3	Application	7	26(52%)
4	Analysis	5	18(36%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		15	50 (100%)